

Questions

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Q1 - 2024 (05 Apr Shift 2)

A vernier callipers has 20 divisions on the vernier scale, which coincides with 19th division on the main scale. The least count of the instrument is 0.1 mm. One main scale division is equal to _____ mm.

- (1) 0.5
- (2) 2
- (3) 5
- (4) 1

Q2 - 2024 (06 Apr Shift 1)

While measuring diameter of wire using screw gauge the following readings were noted. Main scale reading is 1 mm and circular scale reading is equal to 42 divisions. Pitch of screw gauge is 1 mm and it has 100 divisions on circular scale. The diameter of the wire is $\frac{x}{50}$ mm. The value of x is :

- (1) 21
- (2) 142
- (3) 71
- (4) 42

Q3 - 2024 (06 Apr Shift 2)

In a vernier calliper, when both jaws touch each other, zero of the vernier scale shifts towards left and its 4th division coincides exactly with a certain division on main scale. If 50 vernier scale divisions equal to 49 main scale divisions and zero error in the instrument is 0.04 mm then how many main scale divisions are there in

1 cm ?

- (1) 10
- (2) 5
- (3) 20
- (4) 40

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Q4 - 2024 (08 Apr Shift 1)

The diameter of a sphere is measured using a vernier caliper whose 9 divisions of main scale are equal to 10 divisions of vernier scale. The shortest division on the main scale is equal to 1 mm. The main scale reading is 2 cm and second division of vernier scale coincides with a division on main scale. If mass of the sphere is 8.635 g, the density of the sphere is:

- (1) 2.0 g/cm^3
- (2) 1.7 g/cm^3
- (3) 2.2 g/cm^3
- (4) 2.5 g/cm^3

Q5 - 2024 (08 Apr Shift 2)

Least count of a vernier caliper is $\frac{1}{20 \text{ N}}$ cm. The value of one division on the main scale is 1 mm. Then the number of divisions of main scale that coincide with N divisions of vernier scale is :

- (1) $(2 \text{ N} - 1)$
- (2) $\left(\frac{2 \text{ N} - 1}{2 \text{ N}}\right)$
- (3) $\left(\frac{2 \text{ N} - 1}{2}\right)$
- (4) $\left(\frac{2 \text{ N} - 1}{20 \text{ N}}\right)$

Q6 - 2024 (08 Apr Shift 2)

There are 100 divisions on the circular scale of a screw gauge of pitch 1 mm. With no measuring quantity in between the jaws, the zero of the circular scale lies 5 divisions below the reference line. The diameter of a wire is then measured using this screw gauge. It is found that 4 linear scale divisions are clearly visible while 60 divisions on circular scale coincide with the reference line. The diameter of the wire is :

- (1) 3.35 mm
- (2) 4.65 mm
- (3) 4.55 mm
- (4) 4.60 mm

Questions

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Q7 - 2024 (09 Apr Shift 1)

One main scale division of a vernier caliper is equal to m units. If n^{th} division of main scale coincides with $(n + 1)^{\text{th}}$ division of vernier scale, the least count of the vernier caliper is :

(1) $\frac{n}{(n+1)}$

(2) $\frac{1}{(n+1)}$

(3) $\frac{m}{(n+1)}$

(4) $\frac{m}{n(n+1)}$

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Answer Key

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Solutions

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Q1

$$20\text{VSD} = 19\text{MSD}$$

$$1\text{VSD} = \frac{19}{20}\text{MSD}$$

$$\text{L.C.} = 1\text{MSD} - 1\text{VSD}$$

$$0.1 \text{ mm} = 1\text{MSD} - \frac{19}{20}\text{MSD}$$

$$0.1 = \frac{1}{20}\text{MSD}$$

$$1\text{MSD} = 2 \text{ mm}$$

Q2

$$\text{MSR} = 1 \text{ mm}, \text{CSR} = 42, \text{pitch} = 1 \text{ mm}$$

$$\text{LC} = \frac{\text{pitch}}{\text{No. of CSD}} = \left(\frac{1}{100} \right) = 0.01 \text{ mm}$$

$$\text{Diameter} = \text{MSR} + \text{LC} \times \text{CSD}$$

$$\text{Diameter} = 1 + (0.01) \times 42 \text{ mm}$$

$$\text{Diameter} = 1.42 \text{ mm} = \frac{x}{50}$$

$$\therefore x = 71$$

Q3

$$50\text{VSD} = 49\text{MSD}$$

$$1\text{VSD} = \frac{49}{50}\text{MSD}$$

$$\text{LC} = 1\text{MSD} - 1\text{VSD}$$

$$\text{LC} = 1\text{MSD} - \frac{49}{50}\text{MSD} = \frac{\text{MSD}}{50} \dots\dots\dots (1)$$

$$\text{Now, } |\text{zero error}| = |0 - 4(\text{L.C})|$$

$$0.04 = |-4(\text{L.C})| \Rightarrow \text{L.C} = 0.01 \text{ mm}$$

$$0.04 = |-4(\text{L.C})| \Rightarrow \text{L.C} \dots\dots\dots (2)$$

$$= 0.01 \text{ mm}$$

Using (1) and (2)

$$\frac{\text{MSD}}{50} = 0.01$$

$$\text{MSD} = 0.5 \text{ mm}$$

$$\text{MSD} = \frac{1 \text{ cm}}{N}$$

$$N = \frac{1 \text{ cm}}{0.5 \text{ mm}} = \frac{100 \text{ mm}}{5 \text{ mm}}$$

$$N = 20$$

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Solutions

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Q4

$$\text{Given } 9\text{MSD} = 10\text{VSD}$$

$$\text{mass} = 8.635 \text{ g}$$

$$\text{LC} = 1\text{MSD} - 1\text{VSD}$$

$$\text{LC} = 1\text{MSD} - \frac{9}{10}\text{MSD}$$

$$\text{LC} = \frac{1}{10}\text{MSD}$$

$$\text{LC} = 0.01 \text{ cm}$$

$$\text{Reading of diameter} = \text{MSR} + \text{LC} \times \text{VSR}$$

$$= 2 \text{ cm} + (0.01) \times (2)$$

$$= 2.02 \text{ cm}$$

$$\text{Volume of sphere} = \frac{4}{3}\pi\left(\frac{d}{2}\right)^3 = \frac{4}{3}\pi\left(\frac{2.02}{2}\right)^3$$

$$= 4.32 \text{ cm}^3$$

$$\text{Density} = \frac{\text{mass}}{\text{volume}} = \frac{8.635}{4.32} = 1.998 \sim 2.00 \text{ g}$$

SECTION -B

Q5

$$\text{Least count of vernier calipers} = \frac{1}{20 N} \text{ cm}$$

$$\therefore \text{Least count} = 1\text{MSD} - 1\text{VSD}$$

let x no. of divisions of main scale coincides with N division of vernier scale, then

$$1\text{VSD} = \frac{x \times 1 \text{ mm}}{N}$$

$$\therefore \frac{1}{20 N} \text{ cm} = 1 \text{ mm} - \frac{x \times 1 \text{ mm}}{N}$$

$$\frac{1}{2 N} \text{ mm} = 1 \text{ mm} - \frac{x}{N} \text{ mm}$$

$$x = \left(1 - \frac{1}{2 N}\right) N$$

$$x = \frac{2 N - 1}{2}$$

Q6

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Solutions

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$$\text{Least count} = \frac{1}{100} \text{ mm} = 0.01 \text{ mm}$$

$$\text{zero error} = +0.05 \text{ mm}$$

$$\text{Reading} = 4 \times 1 \text{ mm} + 60 \times 0.01 \text{ mm} - 0.05 \text{ mm} \\ = 4.55 \text{ mm}$$

Q7

$$n\text{MSD} = (n+1)\text{VSD}$$

$$\Rightarrow 1\text{VSD} = \frac{n}{n+1}\text{MSD}$$

$$L \cdot C = 1\text{MSD} - 1\text{VSD}$$

$$L \cdot C = m - m \left(\frac{n}{n+1} \right)$$

$$L \cdot C = m \left(\frac{n+1-n}{n+1} \right)$$

$$\Rightarrow L \cdot C = \left(\frac{m}{n+1} \right)$$

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